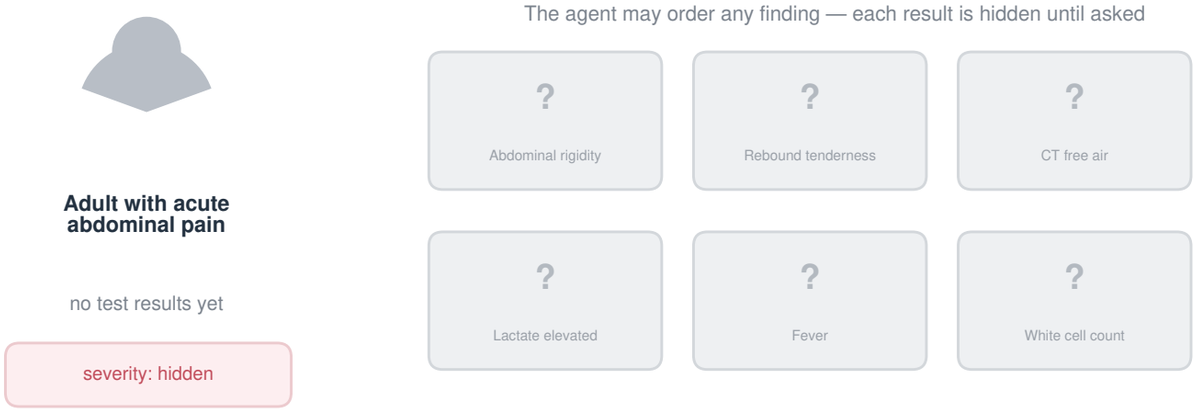
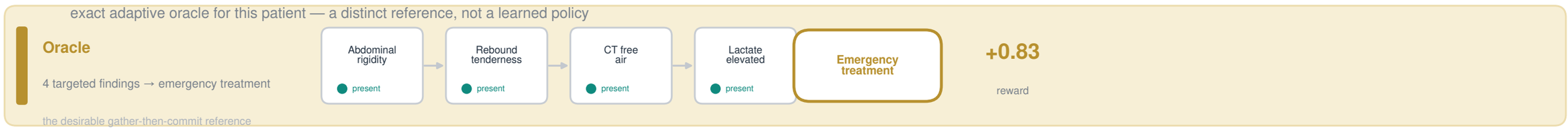
Reward Design Governs Information Gathering in Clinical Multi-Turn Agents

The same patient yields different multi-turn procedures under different reward designs — and the procedure matters apart from the final score.

A The multi-turn clinical problem



B Reference adaptive procedure



C What different reward designs learn

D Reward geometry, and how the procedure shifts

small quantitative anchor — diagnostics, held-out, pure verifier (k = 12)

Refill triage

Continue, modify, hold, or escalate a medication?

Diagnostics · running example

Gather enough evidence to safely disposition an acute presentation

Titration

Choose treatment intensity while balancing competing risks